

ALEX LONSDALE

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PROFILE

MEng Mechanical Engineering student at Lancaster University (First Class to date) specialising in control engineering. Experience designing and comparing control systems in MATLAB and Python, working as part of a team to deliver a working robot, and building automation tools in a live commercial software environment. Seeking a graduate role in control systems, fluid dynamics or simulation and software engineering.

EDUCATION

Lancaster University

Sep 2023 – Jun 2027 (expected)

MEng Mechanical Engineering (integrated master's)

- First Class result to date (Years 1–3); final classification awarded on completion of Year 4
- Specialising in control engineering; coursework includes fluid dynamics with ANSYS

PROJECTS

Individual MEng Project: Automatic Warfarin Dose Management Using Model-Based Control Systems Sep 2025 – Mar 2026

- Reviewed the literature on automated drug dosage control to choose the approach for the project
- Identified transfer functions representing patient metabolic response to warfarin from data, using numerical analysis
- Developed and compared several control systems on these models, and assessed the real-world challenges of implementing them
- Evaluated results and drew conclusions using Python and MATLAB; reflected on own performance, including time management and supervisor feedback

Autonomous Line-Following Robot (team of five)

Sep 2024 – Mar 2025

- Designed and built a robot that follows a black line and fires projectiles at two different targets, using LDRs, microcontrollers and servo motors
- Developed initial concepts as a team and delivered a working prototype to deadline, using planning and organisation tools to stay on schedule
- Completed the course well under the time limit and fired both projectiles on the first attempt; scored very highly

WORK EXPERIENCE

Development Assistant, Poole Alcock Solicitors, Nantwich

Jul 2026 – Sep 2026

- Developed tools on the Proclaim case management system to help case handlers work faster and manage risk more consistently
- Designed and implemented automated risk assessment calculations and escalation rules so high-risk cases were flagged and routed without manual checking
- Replicated and improved existing workflows to cut repetitive work; met stakeholders to shape them around how case handlers actually work
- Tested and iterated changes in a live environment, balancing new functionality against the stability of a system the firm relies on daily
- Trained in Access Group Proclaim and certified in the Technical modules

Digital Marketing Assistant, Poole Alcock Solicitors, Nantwich

Jun 2024 – Sep 2024

- Supported the firm's digital marketing strategy through SEO changes to the website and keyword research
- Wrote copy for new web pages and published content online

TECHNICAL SKILLS

Programming and modelling: Python, MATLAB, Simulink

Engineering: Control system design and comparison, system identification, numerical analysis, fluid dynamics fundamentals

Software and tools: ANSYS, Access Group Proclaim (Technical modules certified), SEO and web publishing

CORE SKILLS

- Effective in teams, with strong organisation and time management skills. My degree and work experience have given me a good foundation to develop these key skills throughout my engineering career
- Hardworking, with a real passion for research and problem solving
- Communicate technical work clearly and listen to the people using it, from shaping workflows around case handlers' needs to writing up and evaluating project findings
- Self-directed: managed a year-long individual research project from literature review to conclusions